Robust Lidar–Radar Composite Cloud Boundary Detection Method With Rainfall Pixels Removal

Weijie Zou, Zhenping Yin, Yaru Dai, Yubao Chen, Zhichao Bu, Siwei Li, Yun He, Xiuqing Hu, Detlef Müller, Tong Lu, Xiangyu Dong, and Xuan Wang

***Abstract*— Cloud vertical structure detection is essential for understanding atmospheric dynamics. Currently, cloud boundaries can be effectively identified based on lidar and millimeter-wave radar. However, how to integrate the two observation methods and remove the interference of rainfall on cloud identification are crucial for precise detection of cloud boundaries. This study develops a robust cloud boundary detection method combining radar and lidar observations with ability to identify rainfall effectively. Consistency analysis at Sheyang meteorological station using radiosonde data showed that lidar detected 41.1% of clouds and radar detected 93.3% of clouds compared to composite detection. The composite method overestimated the cloud base by 855.1 m and underestimated the cloud top by 551.2 m compared to radiosonde, as radiosonde measurements are affected not only by drift but also by rainfall, which mainly affects cloud base detection. Utilizing Doppler velocity and the lidar–radar cloud base difference improved rainfall detection by 41.4% over Doppler velocity alone. Observations are consistent with ground-based rain gauge, with Doppler velocities providing good identification of significant rainfall. Also, different cloud bases detected by lidar and radar providing additional identification of drizzle. With the rainfall removed, the error of rainwater path offered by microwave radiometry during rainfall is reduced by 32.4%. Overall, this study proposes a threshold-insensitive lidar–radar composite cloud identification method. It has good robustness and more precise detection of cloud boundaries for its ability to identify vertical rainfall regions.**

***Index Terms*— Cloud identification, lidar, precipitation, radar, remote sensing.**

I. Introduction

CLOUD is an important part of the Earth's water cycle and radiative transfer [1]. However, their high variability and complex physical processes pose significant challenges to their simulations, necessitating observational methods to elucidate their physical processes and reduce simulation errors [2], [3]. Clouds participate in the global water cycle through formation, development, movement, and dissipation. The vertical structures of primary meteorological factors control these physical processes of clouds and lead to fascinating cloud fields in the horizontal and vertical directions [4], [5], [6]. Chi et al. [7] discovered that the vertical structure of clouds is closely related to seasons and regions through analysis of the global vertical cloud structure, further revealing the interaction between cloud vertical structure and climate [8], making their vertical distribution a key scientific concern. These days, the performance of deep learning models has greatly improved by incorporating vertical descriptions of meteorological factors, even surpassing traditional numerical prediction models [9], [10]. This also highlights the significant importance of vertical observations in atmospheric research. As essential atmospheric components, studying clouds' vertical distribution is therefore scientifically important [11]. The identification of cloud layer vertical boundaries serves as a crucial prerequisite for cloud microphysical parameter inversion and understanding the spatial structure changes of clouds [4], [11], [12], [13].

Since the 20th century, the long-range detection capabilities of millimeter-wave cloud radar (MMCR) and lidar have been widely applied in the observation of cloud vertical structures, enabling the quantitative observation of the vertical spatial distribution of clouds, the interaction between clouds and aerosols, and meteorological processes such as precipitation formation [14], [15]. Cloud identification with lidar commonly uses thresholding [16] or zero-crossing [17] method, but these methods are very sensitive to the setting of thresholds. Zhao et al.'s value distribution equalization (VED) algorithm [18] is threshold-insensitive, not requiring calibrated signals, thus features with improved reliability. However, for extinction coefficients in clouds up to 131 km^{-1} in 532 nm [19], lidar signals attenuate rapidly within them, preventing penetrating clouds with optical depths above 3 [20]. Thus, thick or multilayered clouds challenge lidar identification algorithm due to intense attenuation. Heavy rain can also influence algorithms [21]. However, the research by Li et al. [22] shows that vertical overlapping of clouds is critically important for predicting total cloud fraction (TCF) and radiation budget; therefore, lidar is not sufficient for observing the vertical structure of clouds.

In contrast, radar with longer wavelengths can easily penetrate dense clouds [23], detecting multilayer and thick

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clouds. Radar algorithms typically treat cloud identification as discriminating significant signals above the atmospheric background [24], [25]. These simple and threshold-insensitive algorithms are proven to be reliable, though insensitive to optical thin cloud due to the long wavelengths [26]. They cannot separate precipitation and cloud bases well with radar reflectivity only, leading to the underestimation of cloud bases in rain [13], [27].

Failing to identify rainfall can substantially affect the estimation of liquid water content and rainwater path predictions obtained from microwave radiometers [28], [29] because different assumptions are often made about the particle size distribution when retrieving water content. Rainfall and cloud also have different characteristics in heat flux and radiative effects [2], [30], thus making accurately identifying precipitation regions important.

In clouds without rainfall, particles primarily exist in suspension, with relatively small terminal velocities. Under light rain conditions, vertical air motion hinders the differentiation of light rain [31], referred to as "drizzle" [32]. As shown by Irbah et al. [33], the observation of Doppler velocities is significantly influenced by vertical air currents due to the small particle sizes, leading to greater uncertainty and making it challenging to distinguish drizzle from cloud based on Doppler velocities alone. However, Yi et al. [34] showed that lidar observes microphysical processes at the melting layer, penetrating light drizzle, showing that lidar possesses the ability to observe cloud base under light rain. In heavier rainfall condition, rapid signal attenuation hampers lidar cloud boundary identification. However, radar spectral width and Doppler velocity can also indicate precipitation [31], [35], [36]; thus, both lidar and radar have partial capabilities and limitations in identifying precipitation. After raindrops form, there is a significant difference in particle size between cloud droplets and raindrops [37]. As mentioned in Beard's study, the typical size of cloud droplets is less than 20 μm, while raindrops range from 1 to 7 mm [38]. Because the number concentration of rain is much lower than that of cloud droplets, the lidar signal changes markedly, and the Doppler velocity increases significantly as well for the increasing particle size. This characteristic shows the potential for cloud and rain classification. Following [26], only liquid precipitation is considered in this research.

Lidar and radar have distinct strengths for cloud boundary and precipitation identification.

To fully exploit both instruments, Venema et al. [13] proposed a composite approach in 2000. This approach uses lidar for cloud base detection and radar for cloud top detection while handling precipitation separately. Bühl et al. [39] combined lidar and radar for phase and precipitation identification but did not fully utilize lidar in cloud base detection and not discriminate drizzle. Garrett and Zhao conducted joint observations using a ceilometer and cloud radar in Antarctica. They effectively utilized the cloud base information obtained from the ceilometer and the reflectivity factor acquired from the cloud radar to estimate the precipitation phase and rainfall amount and attempted to further differentiate between clouds and rain [27]. In addition, they performed analysis on thin clouds [40], making significant contributions to the coordinate

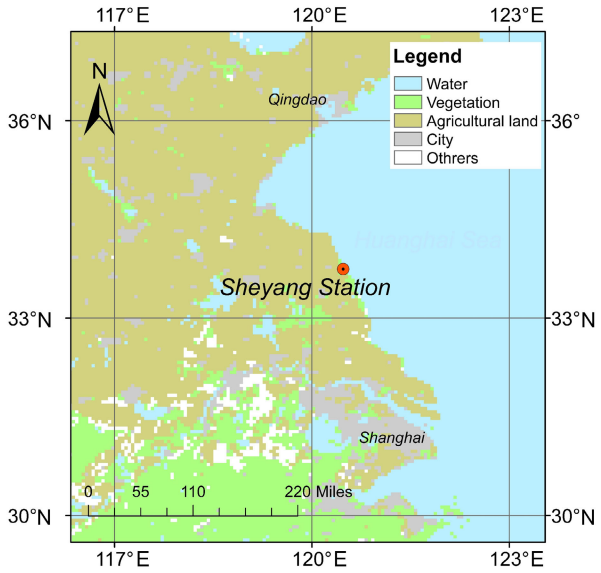


Fig. 1. Geographic location of Sheyang meteorological observation station.

observations of lidar and cloud radar. Yuan et al. [41] analyzed individual and joint capabilities of radar–lidar cloud detection, taking radar detection data as the main data and lidar data as the assistant data. Yuan et al. also came out with an observation guideline for different detection conditions such as drizzle. Okamoto et al. [12] utilized threshold-based cloud detection algorithms for different observation instruments, merging cloud masks according to various conditions, and demonstrated an application case over the ocean. However, the cloud detection scheme employed is sensitive to threshold values, and the potential for improving rainfall identification using composite observations has not been fully exploited.

Building on these studies, this article presents an enhanced composite method for robust cloud boundary identification and improved rainfall delineation. Consistency with regular radiosonde data from China Meteorological Administration (CMA) is analyzed. Case studies and time series are given to assess the rationality and reliability.

II. Data and Instrumentation

The observation station is located in Sheyang County, Jiangsu (see Fig. 1). This study uses observation data from this station as an example to validate the effectiveness of the proposed method. Sheyang is situated adjacent to the Huanghai Sea with flat terrain at an altitude near sea level. The area has abundant agricultural resources and lush vegetation cover near the site, which features with large amount of water vapor resources and cloud cover.

A three-wavelength lidar and MMCR have been deployed at Sheyang station to monitor the atmospheric environment. Regular radiosonde launches are also performed by the station. Due to substantial water vapor influx from the nearby sea and local transpiration, Sheyang experiences a high probability of precipitation. The area is also susceptible to typhoons and tornadoes. Therefore, it was selected as the validation site for the enhanced cloud detection algorithm with improving rainfall identification. This study analyzes data from July to October

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2023, utilizing MMCR, lidar, and radiosonde observations for cloud detection algorithm development and verification.

The data used are uncalibrated raw data. The routine lidar calibration applied in the lidar network relies on Rayleigh fit results, in which the lidar profiles with clouds need to be removed. Cloud detection is a crucial part of the processing flow from lidar raw data to calibrated data, which explains why uncalibrated data are used for cloud detection.

A. 35-GHz Millimeter-Wave Cloud Radar

The MMCR data utilized in this study were acquired from the ground-based vertical remote sensing observation system operated by the CMA. The radar operates at a Ka-band frequency pointing vertically. It has a range resolution of 30 m and an effective detection range of 15 km. The MMCR has a peak transmit power exceeding 20 W and can measure radar reflectivity factor, Doppler spectral width, and Doppler velocity. The reflectivity factor measurement range is from −40 to 30 dBZ within 1 dBZ. The velocity measurement range is from −15 to 15 m/s with the error of no more than 0.5 m/s. The spectral width can be measured from 0 to 15 m/s within 0.5 m/s. The temporal resolution is 1 min.

B. Lidar

The lidar data utilized in this study were collected by the three-wavelength lidar operated at Sheyang Detection Center of the CMA. This lidar emits pulses at wavelengths of 355 nm with a max pulse energy of 69 mJ, 532 nm with a max pulse energy of 188 mJ, and 1064 nm with a max pulse energy of 141 mJ. The range resolution is 15 m in vertical. The lidar has an incomplete overlap from the ground to 500 m. The operating frequency is 1 kHz and 1-min temporal resolution, allowing continuous observations throughout the day. This work analyzes signals from the parallel 532-nm channel, which is a direct measure of backscatter signal from aerosols and hydrometeors.

C. Radiosonde Data

Radiosonde data were collected twice daily at the local time of 8:00 and 20:00 (UTC+8), with the balloon launch site situated approximately 60 m from the radar, ensuring co-located observations. The sensors carried by the radiosonde measure temperature, relative humidity, and air pressure from the surface up to 16-km altitude. The average ascent rate of the balloon carrying the radiosonde is around 400 m/min. The radiosonde samples at a period of 1.2 s, providing a vertical spatial resolution of 8 m in the resulting data profiles.

III. Methodology

A. Lidar–Radar Composite Cloud Boundary Detection Scheme

It is generally believed that lidar and radar have different characteristics in cloud boundary detection tasks [36], [42]. Therefore, to effectively utilize the cloud boundary detection advantages of each, a reasonable fusion method is required for cloud boundary detection tasks.

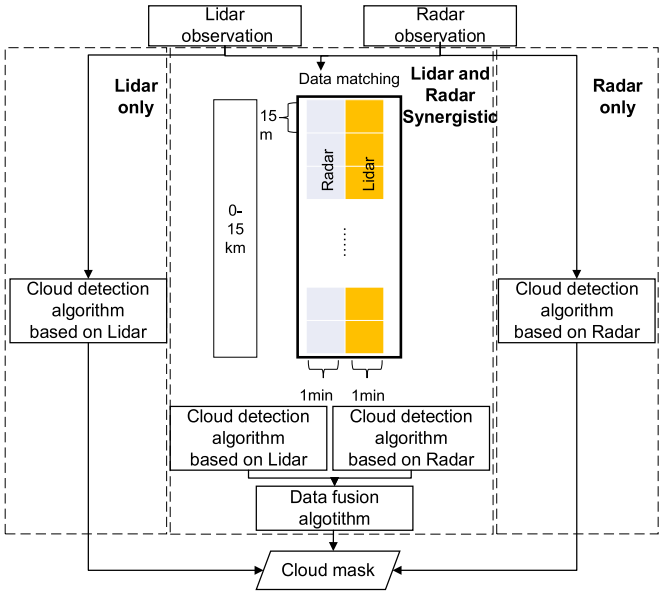


Fig. 2. Schematic of composite cloud boundary detection algorithm.

The overall process is outlined in Fig. 2. This study accounts for potential inconsistencies between the lidar and radar datasets. The composite cloud boundary detection algorithm can output valid results even with only one data type available. When both lidar and cloud radar observations are present, the data are first spatially and temporally matched.

To integrate cloud observations from different instruments, common approaches include retrieving physical parameters from signals to identify clouds or performing cloud identification separately from each instrument's observations and then merging the results. This study adopts the latter approach. Cloud possibility is used to describe the likelihood of clouds in a given space, replacing simple cloud masks. This possibility ranges from 0 to 1, indicating the possibility of cloud presence from clear to fully cloudy. This approach was chosen for two reasons. First, it retains the method's scalability, allowing the inclusion of new observation techniques. Second, direct inversion of physical parameters from signals involves numerous uncertainties. Therefore, this study employs a phenomenon-based fusion approach.

The cloud identification algorithms for the individual instruments are detailed in Sections III-B and III-C. Although the fusion method builds on the single-sensor-based identification, necessary data normalization beforehand and more rigorous postprocessing after hand are performed. The fusion methodology is elaborated further in Section III-D.

B. Cloud Boundary Detection Based on Lidar Only

Lidar signals contain contributions from atmospheric background radiation and instrument noise in addition to backscatter from atmospheric constituents. To account for this, the mean signal from the far range (25–30 km) is assumed to represent the background and noise level. As shown in Fig. 3, subtracting this average value removes these extraneous contributions and smoothing is also used to filter noise, completing the data preprocessing.

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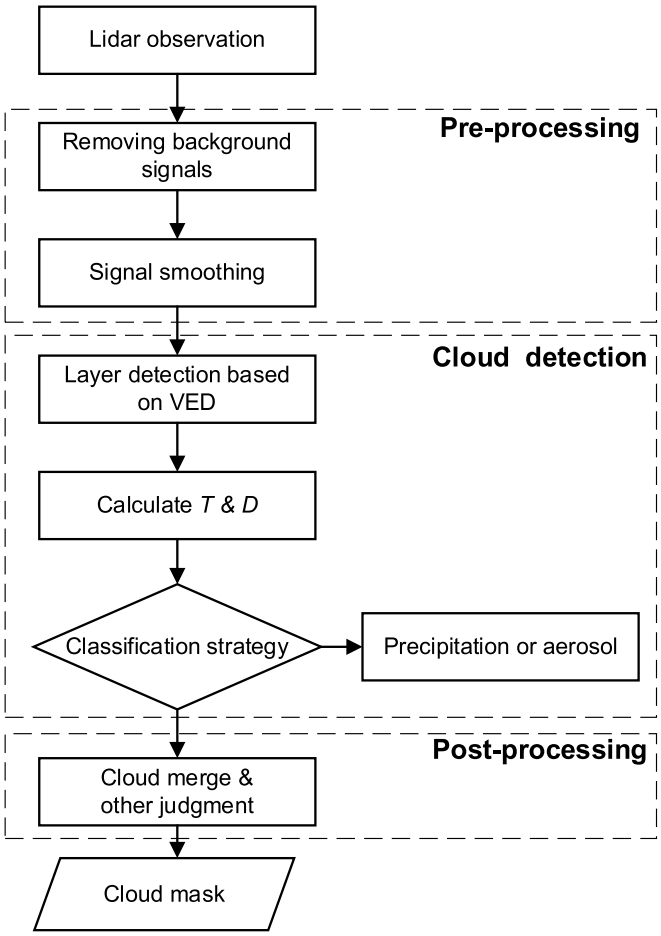


Fig. 3. Schematic of cloud boundary detection algorithm based on lidar-only observation, where T means the signal gradient and D means the attenuation rate.

Lidar-based cloud identification relies on recognizing features layer in the lidar profiles. Layer detection mainly uses the VED algorithm proposed by Zhao et al. [18], which can effectively identify cloud layers from uncalibrated signal profiles and is thus adopted here. The VED algorithm sorts the lidar signal based on equalization to delineate anomalous gradient features corresponding to layer boundaries.

Subsequently, this study assigns different cloud possibilities based on the attenuation rate (D) and signal gradient (T) of the extracted feature layers. Generally, T within a cloud layer is significantly larger and D is much smaller than in aerosol regions, allowing empirical thresholds to distinguish between the two. These parameters are also widely used in other cloud identification studies [17], [43], [44]. Instead of simply identifying clouds in a binary mask, this approach transforms it into a probability. The classic threshold values are used as reference midpoints [18], and by setting upper and lower limits, the calculation of cloud possibilities is linearized. This makes the identification algorithm robust to a range of T and D values, assigning cloud possibility linearly based on their magnitudes.

As shown in Table I, this study adopts the thresholds proposed by Zhao et al. [18] as a midpoint. For cloud height (CH) below 3 km, the midpoints are set at $T = 3$ and $D = -7$. Z, As shown in formulas (1) and (2), when T is between 2 and 4, the cloud possibility ranges from 0 to 1; similarly, when D is between -4 and -10, the cloud possibility also ranges from 0 to 1. Values below the minimum threshold are assigned a possibility of 0, and values above the maximum threshold are assigned a possibility of 1

TABLE I
DIFFERENT SIGNAL GRADIENT T AND ATTENUATION RATE D TO MINIMUM CLOUD POSSIBILITY AND MAXIMUM CLOUD POSSIBILITY

	Cloud possibility = 0	Cloud possibility = 0.5	Cloud possibility = 1
Cloud base height < 3km	T = 2 & D = -4	T = 3 & D = -7	T = 4 & D = -10
Cloud base height >= 3km	T = 0.5 & D = -2	T = 1.5 & D = -5	T = 2.5 & D = -8

$$\text{Cloud possibility}_T = \begin{cases} 0, & \text{when } T < 2 \\ \dfrac{T-2}{2}, & \text{when } 2 < T < 4 \\ 1, & \text{when } 4 < T \end{cases} \quad \text{When CH} < 3 \text{ km} \tag{1}$$

$$\text{Cloud possibility}_D = \begin{cases} 0, & \text{when } -4 < D \\ \dfrac{D+4}{-6}, & \text{when } -10 < D < -4 \\ 1, & \text{when } D < -10 \end{cases} \quad \text{When CH} < 3 \text{ km.} \tag{2}$$

As shown in formula (3), the overall cloud possibility from lidar observation is determined by averaging the cloud possibility from T to D, assuming equal weights

$$\text{Cloud possibility}_{\text{lidar}} = W_T \times \text{Cloud possibility}_T + W_D \times \text{Cloud possibility}_D \tag{3}$$

where W_T is 0.5 and W_D is 0.5 in this study.

In the most classical scenario where $T = 3$ and $D = -7$, the total cloud possibility is 0.5. Therefore, layers with possibilities greater than or equal to 0.5 are considered highly likely to be clouds. Since the distribution of clouds and aerosol layers is closely related to their altitude [45], this algorithm is also adjusted according to altitude.

Although this method uses thresholds for processing T and D, these thresholds are derived from reliable [17], [43], [44] and linearization considers the uncertainties, thus providing robust results. The threshold for cloud possibility that should be used for final identification will be discussed in detail in case 1.

Therefore, this approach remains applicable even with uncorrectable attenuation in precipitation and is not threshold sensitive. Moreover, all analyses are performed on individual profiles. It is worth noting that rainfall can attenuate the lidar

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signal, reducing its cloud detection capability. The discussion involving rainfall will be further elaborated after.

As lidar signals propagate through the atmosphere, they are attenuated gradually. When the signal intensity is attenuated to a certain degree, cloud boundary cannot be identified reliably. To assess signal strength, the signal-to-noise ratio (SNR) is defined as:

$$\mathrm{SNR} = \frac{P_m(z) - P_{\mathrm{bg}}}{\sqrt{P_m(z)}} \tag{4}$$

where P_{bg} is the background noise signal, which is obtained by averaging the raw signal at far ranges from 25 to 30 km, and $P_m(z)$ is the signal power at height z. In the analysis, the cloud possibility is adjusted based on the SNR. When the SNR drops below 3, the detected cloud top is considered unreliable, but the cloud base remains reliable. The cloud possibility decreases linearly from the range bin where SNR $>$ 3 below the cloud top, reaching 0 at the cloud top itself.

For cloud boundary based on lidar detection, it is identified when the cloud possibility exceeds 0.5, which is finally denoted by a cloud mask. Postprocessing involves checking minimum cloud thickness and cloud base height. According to Xu et al. [29], the minimum cloud geometrical thickness (CGT) is set as 30 m, which is twice the vertical resolution of the lidar, as thinner cloud layers are likely to be caused by noise or other artifacts. The lidar has an incomplete overlap from the ground to 500 m, which hinders the cloud detection below this height. Therefore, the minimum cloud base is set to 500 m.

Importantly, this single-sensor postprocessing is not used in the fusion algorithm. Unified postprocessing is done after merging the data.

C. Cloud Boundary Detection Based on Radar Only

As shown in Fig. 4, radar-based cloud identification starts with a clear-air echo filter, which impedes cloud boundary detection near the ground. The spectral width and Doppler velocity are used to filter clear-air echo in this method.

Following [46], clear-air echo is filtered when the spectral width $>$0.3 m/s and velocity $<$0.2 m/s. Then, cloud is identified as discerning whether the signal at a range gate significantly deviates from the clear-air profile. The cloud layer is determined using the 3σ principle, compared to the mean and standard deviation (σ) of the profile. Different cloud possibilities are determined based on the standard deviation sigma level exceeded.

Since at the Ka-band, cloud radar is capable of detecting echoes from clouds or precipitation, excluding insects and airborne objects, it is reasonable to consider this as a significant discrimination problem. Similar to the lidar cloud identification scheme, the mean radar reflectivity of the profile corresponds to the lowest cloud possibility and the maximum cloud possibility corresponds to three times the standard deviation (3σ). Cloud possibilities are assigned linearly based on the signal magnitude from the mean to 3σ. For example, when the signal is mean plus 1.5σ, the cloud possibility is 0.5.

Precipitation is identified when the Doppler velocity <-4 m/s [35].

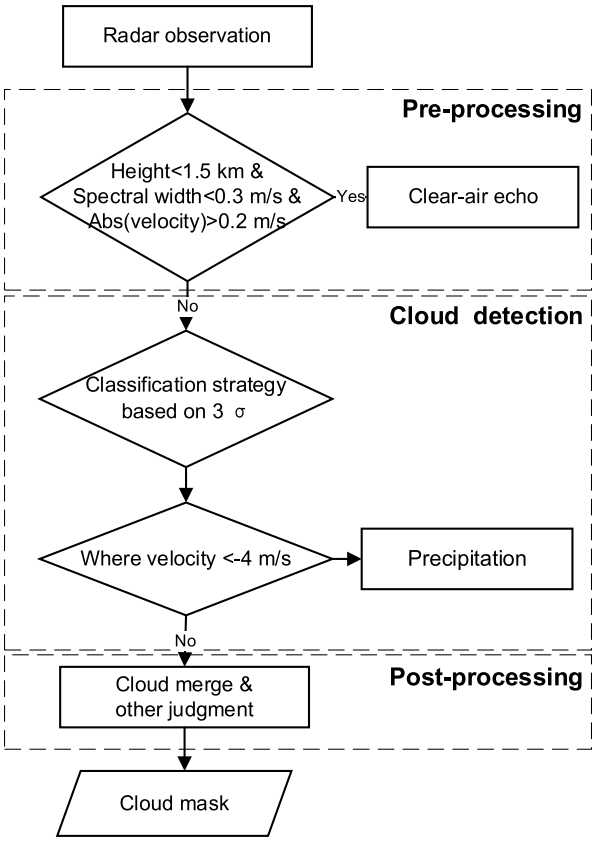


Fig. 4. Schematic of cloud boundary detection algorithm based on radar-only observation.

Similar to the lidar approach, cloud possibilities above 0.5 indicate clouds. The postprocessing same as lidar method is conducted after cloud detection. When composite observations are made, data postprocessing will not be done here but will be done uniformly after the composite cloud boundary detection is complete.

D. Cloud Boundary Detection Based on Composite Observations

Cloud boundaries from different instrumentations and retrievals need to be merged to build up a unified cloud boundary product. Simply merging the lidar and radar cloud masks is unreasonable due to their different detection mechanisms. The fusion approach (see Fig. 5) converts the binary masks into cloud possibility, with merging performed by addition rather than intersection. This enables cloud confirmation through joint detection to avoid the failure of single instrument under certain conditions.

As mentioned, the lidar can penetrate drizzle. Thus, the lowest lidar cloud boundary is used as the reference. Radar signals below this level are identified as potential drizzle. Cloud possibilities are summed by profile and values above 0.5 are labeled as clouds, which will be further discussed in case 1.

After profile fusion, a time–height cross section is generated. Continuity is enforced vertically within profiles, but observation errors can break contiguity. Therefore, frequency filtering with a temporal width of 5 and an occurrence frequency of 2 is applied over time to ensure continuity. Within

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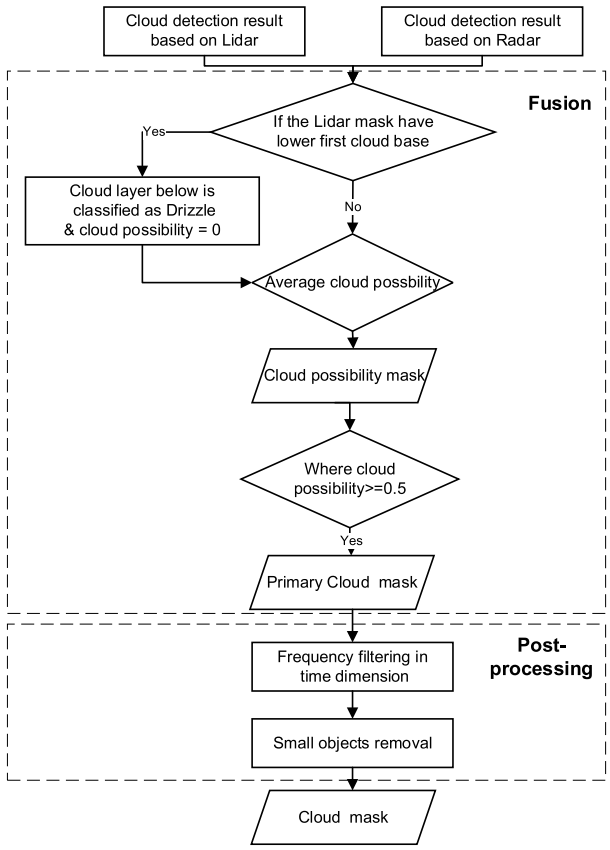


Fig. 5. Schematic of cloud boundary detection algorithm based on combined observation.

five adjacent profiles, if only one pixel at the same height is identified as a cloud, it is considered noise or noncloud factors such as insects. Thus, it is removed.

Finally, very small cloud fragments are treated as noise or wispy vapor. A connected component filter removes small isolated objects in both the clear sky and cloud regions, generating more physically reliable results.

E. Cloud Boundary Detection Based on Radiosonde

Radiosonde is commonly used in situ atmospheric profiling technique [16], [29], [46], [47], [48], [49]. To validate the fusion algorithm, radiosonde cloud identification results are used for comparison.

Radiosondes measure pressure, temperature, and humidity. Following threshold-based cloud identification method WA95 from Wang and Rossow [4], relative humidity is used for cloud boundary detection. However, in this method, the threshold is the same at all altitudes, which is unreasonable. Thus, the improved threshold approach by Zhang et al. [47] is implemented here, threshold of which is a function of altitude. However, Zhang's technique removed thin layers before merging, potentially missing some fragmenting clouds. This order is reversed here.

The specific process mainly includes the following steps.

1) Remove the wet layer with a bottom height less than 120 m and CGT less than 400 m.
2) For wet layer identification, data grid with relative humidity less than the Min-RH (see Table II) is identified as a wet layer.

TABLE II
RELATIVE HUMIDITY THRESHOLD USED IN RADIOSONDES CLOUD IDENTIFICATION REFERRING TO ZHANG [47]

Altitude / km	RH		
	Min-RH	Max-RH	Inter-RH
0 ~ 2	90% ~ 92%	93% ~ 95%	82% ~ 84%
2 ~ 6	88% ~ 90%	90% ~ 93%	78% ~ 80%
6 ~ 12	80% ~ 88%	80% ~ 90%	70% ~ 78%
> 12	75%	80%	70%

3) If the distance between the two layers is less than 300 m, a merging operation is performed.
4) Remove the wet layer with a CGT less than 30 m.
5) The cloud bottom is set to 280 m above the ground, and if the cloud top is lower than 280 m, the layer is removed.

It is important to point out that radiosonde measurements themselves have considerable uncertainty. Comparisons will be conducted further in Section IV.

F. Rainfall Detection

In this study, terminal velocity and particle size are considered primary criteria for distinguishing between clouds and rain [33].

This study explores the vertical rainfall region identification ability through different cloud bases detected by lidar and radar, combined with Doppler velocity. Two steps are used for rainfall identification.

First, Doppler velocities are used to discriminate significant falling raindrops. The first step is based on the fact that after cloud droplets grow into raindrops, they fall at speeds well above the vertical air motion. When the droplet size is less than 10 μm, as Beard [38] shown in the following, the terminal velocity of droplets in cloud can be obtained using the simplified Stokes–Cunningham equation:

$$V_t = C_{\text{sc}}\ \text{Diameter}^2 \tag{5}$$

where V_t means the terminal velocity and C_{sc} means a correction. When the droplet size exceeds 1 mm, cloud droplets become unstable and begin to form small raindrops, while the Reynolds number increases. After rainfall droplets form, their terminal velocities also follow an empirical distribution shown as [50]:

$$V_t = 1.767\ \text{Diameter}^{0.67}. \tag{6}$$

Both cloud and rainfall droplets show a positive relationship between the droplet size and the terminal velocity. Therefore, using a terminal velocity threshold for distinguishing between rainfall and cloud droplets is reasonable. Hence, following [35], Doppler velocities below −4 m/s under 5-km altitude are designated as rainfall. However, as shown in (7), the Doppler velocity does not directly equate to the terminal velocity of liquid droplets, and observed values are influenced by vertical air motions in the atmosphere, introducing a degree of uncertainty

$$V_t = V_D - V_a. \tag{7}$$

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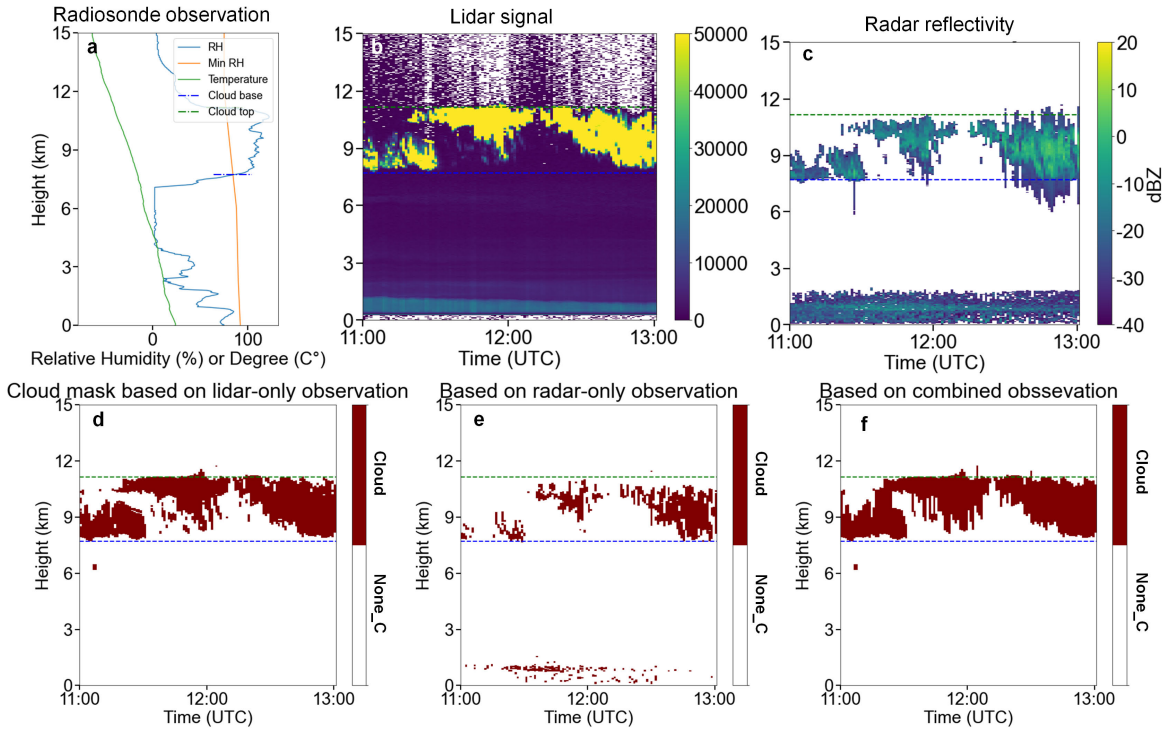


Fig. 6. Comparison with radiosonde cloud detection result. (a) Observation of radiosonde. (b) Lidar signal. (c) Radar reflectivity. (d) Lidar detection result. (e) Radar detection result. (f) Composite detection result.

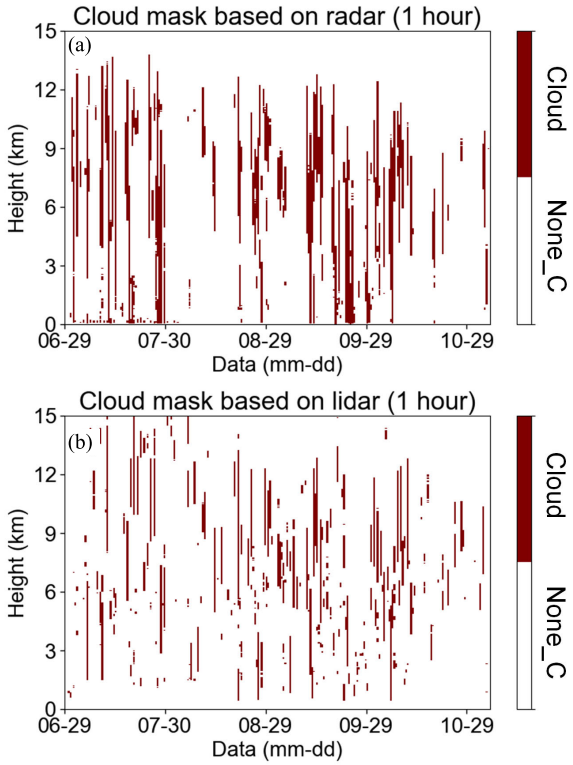


Fig. 7. Cloud mask results obtained by merging radar observations 1 h before and after the release of the radiosonde balloon. At a specific height of a profile, if clouds are identified at any time within this 2 h window, it is considered that clouds are present at that profile. (a) Cloud mask from radar observations. (b) Cloud mask from lidar observations.

Smaller droplets between 20 μm and 1 mm, with lower terminal velocities and low to intermediate Reynolds numbers, are particularly sensitive to air currents and atmospheric disturbances. These droplets are referred to as drizzle in this study. Therefore, in method relying on Doppler velocity observations for rainfall identification, many rainfall areas and cloud areas are misclassified due to the influence of turbulent vertical updrafts, even though their particle sizes are already significantly different. When larger droplets dominate the backscatter signal but have a minimal impact on cloud water content, the corresponding microphysical parameter retrieval schemes can fail. Accurate identification of the cloud base during rainfall processes is crucial for the subsequent retrieval of cloud microphysical parameters such as cloud water content and effective particle size. Hence, more effective identification of rainfall areas can be achieved through combined observations from lidar.

Per the Mie scattering theory, the radar reflectivity can be written as

$$Z = \int N(D) D^6 \mathrm{d}D \tag{8}$$

where N is the number of particles and D is the diameter, with intensity proportional to the particle size to the sixth power [51]. The extinction coefficient can be written as

$$E = C \int N(D) A(D) \mathrm{d}D \tag{9}$$

where $A(D)$ is the area, $A(D) \propto D^2$, and C is a constant. For lidar, the signal is proportional to the square of size [37].

Typical cloud droplets range from 1 to 20 μm, while raindrops span from 0.1 to 7 mm [38]. Thus, raindrop diameters exceed cloud droplets by near two orders of magnitude (10 μm versus 1 mm). Assuming constant water content, the number concentration decreases by nine orders of magnitude. As radar is far more sensitive to the change in raindrops size, the radar signal stays strong and the lidar signal decreases in rainfall areas compared to clouds, which means that lidar can provide reliable cloud base if it is not attenuated. This

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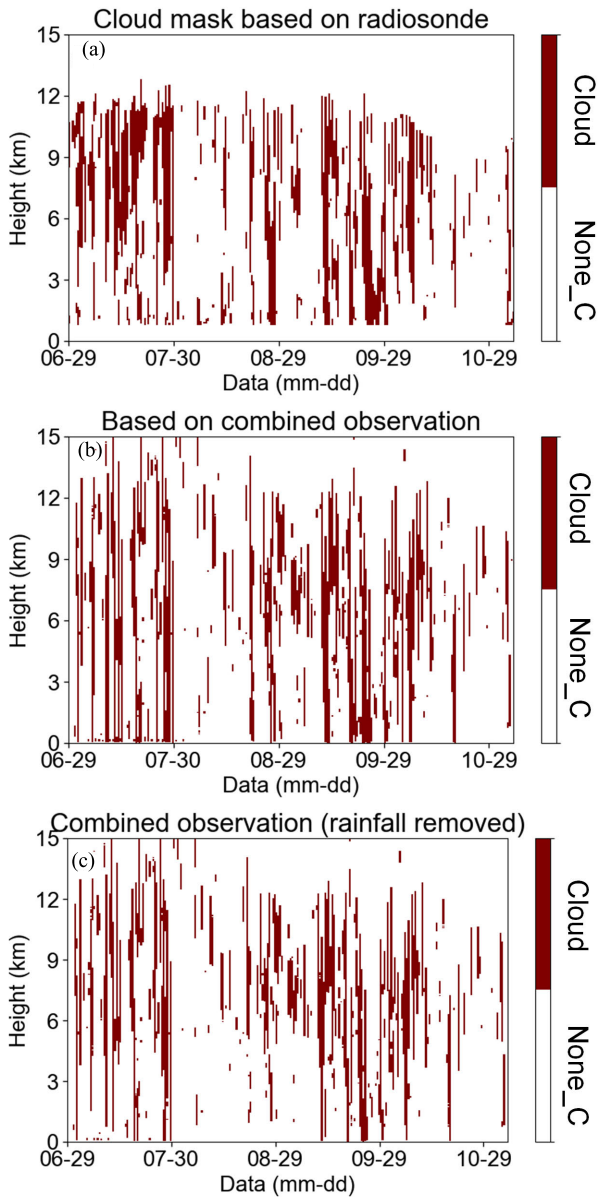


Fig. 8. Time-series cloud mask comparison of (a) radiosonde, (b) lidar–radar observation (with rainfall), and (c) lidar–radar observation (rainfall removed).

difference enables delineating rainfall below the lidar cloud base. However, in most cases, lidar tends to experience rapid attenuation within rain due to either refraction caused by large raindrops or excessive liquid water content. Therefore, under typical circumstances, only drizzle can be reliably detected.

Thus, in the second step, significant radar signals below the lidar cloud base are classified as rainfall. The radar is far more sensitive to the larger raindrops, while lidar signals are dominated by cloud droplets. This difference enables delineating rain below the lidar cloud base while using lidar–radar combined observation.

From (5) and (6), it can be seen that terminal velocity is positively correlated with particle size. Thus, the two parameters are fundamentally consistent. In this article, the cloud base refers to the location where significant changes in terminal velocity and particle size of cloud particles are observed. Assuming continuity in cloud water content variation, (9) indicates that under conditions where the lidar signal is not

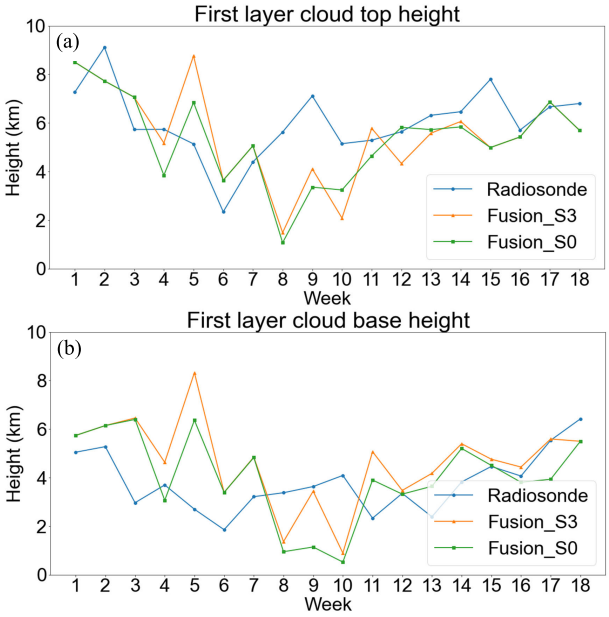


Fig. 9. Comparison in cloud boundary detection. (a) First cloud layer top and (b) first cloud layer base, where S0 means the fusion strategy that does not remove rainfall and S3 means the fusion strategy that removes rainfall.

completely attenuated, the signal from the cloud base is weaker than that from the cloud top but stronger than that from aerosols or the atmosphere. An increase in signal strength suggests a significant increase in particle size and a decrease in number concentration, which aligns with the process of rainfall formation. The stronger signal observed by the cloud radar at the cloud base also indicates a significant increase in particle size. The consistency among Doppler velocity, cloud radar, and lidar signals with particle size changes will be further analyzed in Section IV.

It is worth noting that in this study, precipitation primarily refers to liquid precipitation. Solid precipitation, such as snow, which has optical characteristics similar to ice clouds and exhibits less distinct Doppler spectral distributions [25], is not further differentiated. Only cloud and rainfall are identified.

In this method, bins where both clouds and rain coexist pose a challenge for quantitative description. Distinguishing between them with certainty is difficult due to inherent observational uncertainties. However, early stage rain within these bins may resemble clouds more, exhibiting strong reflectivity while raindrops' terminal velocity remains low. As rainfall intensifies, with distinct Doppler velocities and a decrease in particle concentration, it becomes easier to identify as rain. Quantitative classification schemes for such scenarios will be explored further in the discussion.

Verification and analyses in the results will further evaluate the rainfall identification capability.

IV. Results

A. Validation: Comparison With Time-Series Radiosonde Observations

As this scheme is based on the use of remote sensing techniques, the reliability of the lidar and radar (hardware and data) is assessed through consistency testing against radiosonde observations from the site.

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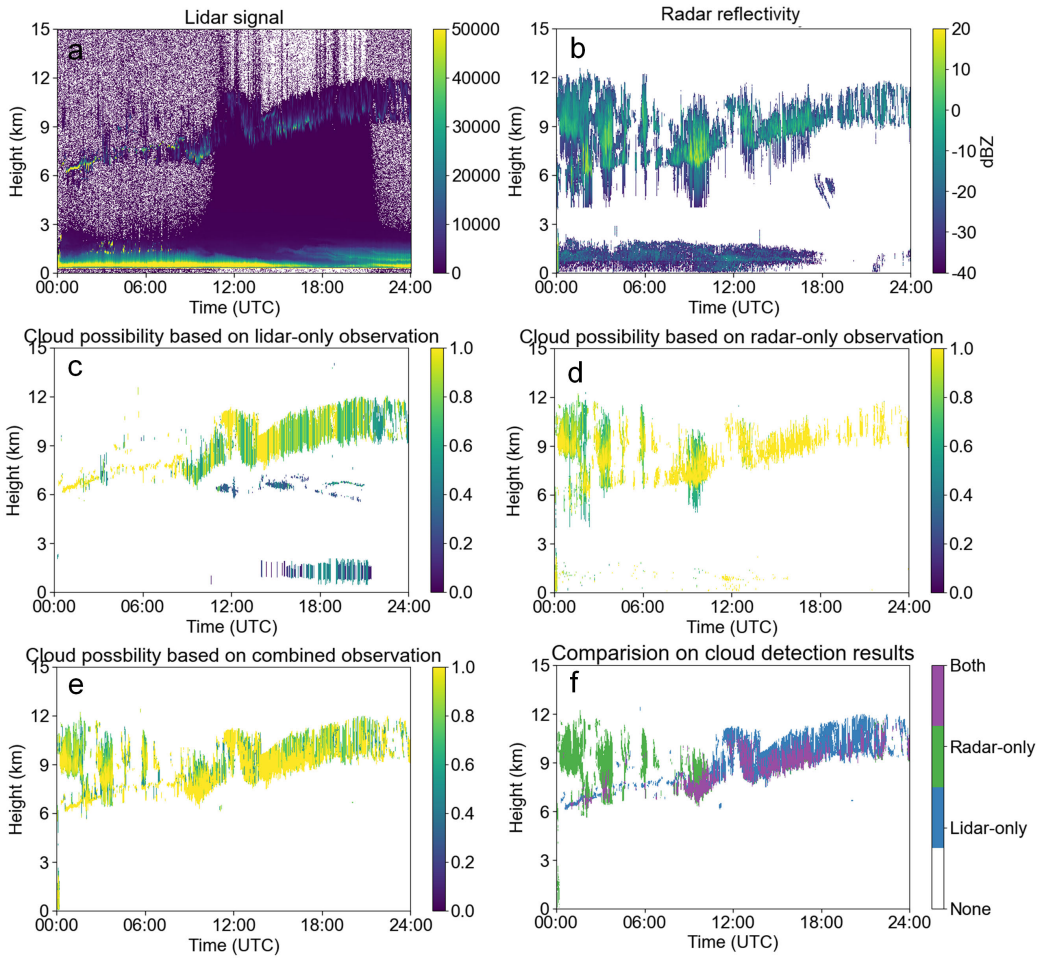


Fig. 10. Observation results on August 30 at the Sheyang station. (a) Lidar signal. (b) Radar reflectivity. (c) Cloud possibility based on lidar cloud detection algorithm. (d) Cloud possibility based on radar cloud detection algorithm. (e) Cloud possibility based on combined cloud detection algorithm. (f) Comparison of the detection result from radar-only, lidar-only, and combined observation algorithm.

A radiosonde launch occurred at 12:00 coordinated universal time (UTC) on September 20 at the observation site. Comparing before and after this time, the 12:00 UTC profiles of lidar and radar differ from the radiosonde identification (see Fig. 6). However, results within an hour of the launch agree well. This discrepancy is likely due to radiosonde balloon drift.

This study also compares the regular release of radiosondes with time-series lidar and radar observation between June 29 and November 5.

As the radiosonde measurements have spatiotemporal drift [46], the radar results at launch time alone are insufficient for comparison. Radar and lidar detection results within 1 h before and after launch time are regarded relevant and are combined in a merge fusion for visualization in Fig. 7.

Fig. 7 shows the radar contains clear-air echo absent in the lidar, and lidar detects some thinner layers. Both radar and lidar are quite different with the detection result from radiosonde [see Fig. 8(a)] and can observe clouds above 12 km ceiling, at which radiosonde fails to detect clouds. However, although single instruments have similar trends in cloud identification as radiosondes, e.g., more clouds in July and fewer clouds in August, there are major differences in the specific identification.

The composite algorithm shows better consistency than the single instrument or radiosonde (see Fig. 8). Before August 1, the composite cloud identification results with rainfall removed are significantly more consistent with the radiosonde results. This indicates that the removal of precipitation areas is indeed important for precise detection of cloud boundaries.

For comparison, the first layer cloud base height and the cloud top height are statistically analyzed (see Fig. 9). The composite algorithm agrees well with the radiosonde observation, and the base height is higher after rain removal [see Fig. 9(b)]. However, weeks 4–6 for cloud base and weeks 8–10 for cloud top still show larger errors compared to the algorithm that keeps rainfall. In weeks 4–6, fewer clouds make statistics more susceptible to systematic and random errors. Radiosonde detects many clouds close to ground, while composite observation did not and high ice clouds exceeded the radiosonde range, contributing disagreement. In weeks 8–10, unfiltered rain likely caused misidentification of thin clouds. Drift and rain impact the radiosonde cloud identification.

In order to obtain a more qualitative evaluation analysis and further intuitively compare the differences between different observation methods, statistical analysis is performed based on the first layer cloud bottom and CH. Comparing cloud base,

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the algorithm overestimates by 855.1 m (21.3%) versus the radiosonde, mainly due to misidentification of precipitation by radio soundings. Without rain removal, the underestimate is only 228.8 m (5.7%), also indicating that the radiosonde cloud detection algorithm used in this study can poorly discriminate rain areas. The reliability of rainfall identification will be further validated through a case study after. For cloud top, the underestimate is 551.2 m (9.2%) versus the radiosonde, potentially due to radiosonde sensor saturation during cloud passage.

B. Case 1: The Cloud Boundary Detection Performance of the Algorithm

Fig. 10 shows the performance of composite algorithm in cloud identification during a measurement performed from 00:00 to 24:00 UTC on August 30 at Sheyang station. From 00:00 to 10:00, the lidar-signal penetration depth sharply declined. Lidar can only detect the cloud base on the lidar-signal attenuation. However, the radar detected cloud layers from 9 to 12 km. From 18:00 to 24:00, the radar failed to delineate the cirrus clouds at 10 km. In contrast, continuous tenuous cirrus clouds can be well identified by lidar measurements. In addition, from 15:00 to 22:00 at 1.5 km, the lidar cloud identification algorithm sensitively detected a characteristic layer. However, due to its low signal attenuation rate and cloud possibility, it was ultimately identified as noncloud, which aligns moderately with the cloud radar observations. Fusing the data provides more continuous cloud identification, demonstrating the value of the composite algorithm.

Simultaneously, this study recommends using 0.5 as the final threshold of cloud possibility for cloud identification. First, a cloud possibility of 0.5 corresponds to the referenced statistical values in the lidar cloud identification algorithm. Second, comparative analysis shows that a threshold of 0.5 is most suitable for practical applications. This is illustrated using the observation data on August 30, as shown in Fig. 11. When the cloud possibility threshold is set to 0.3, aerosol layers are easily misidentified as clouds; when the threshold is set to 0.75, the algorithm tends to miss thin clouds. Overall, a threshold of 0.5 best meets the comprehensive requirements.

The composite algorithm significantly improves cloud detection results compared to the cloud identification algorithm based on single instrument observations. Time-series analysis of date from June 29 to November 5 shows that, in terms of spatial distribution, the pixel-level radar identification results indicate that lidar only detects 41.1% of the clouds compared to the detection results in the fusion data, while the radar detects 93.2% of the clouds. These numbers are influenced by the observation location and seasonal variations. The primary reason is the abundant water vapor during the summer at the experimental site, resulting in many thick clouds that the lidar cannot penetrate.

In addition, the algorithm effectively filters radar clear-air echo and spurious streaks under the cloud bottom. Clear-air echo remains in the single radar observation [see Fig. 10(d)] but is removed after adding lidar detection results [see Fig. 10(e)]. Others are removed primarily by eliminating small isolated patches and frequency filter. Test cases were conducted to validate the effect of frequency filtering and

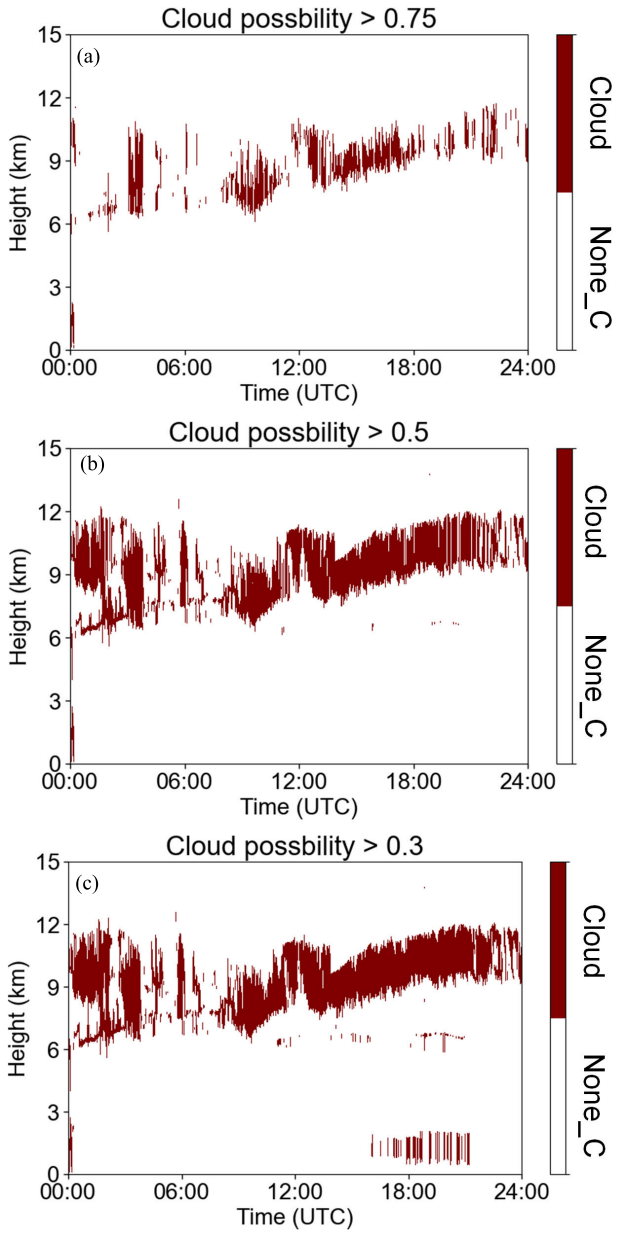


Fig. 11. Comparison of the selection of cloud possibility thresholds. (a) When the cloud possibility is greater than 0.75, it is considered as cloud. (b) When the cloud possibility is greater than 0.5, it is considered as cloud. (c) When the cloud possibility is greater than 0.3, it is considered as cloud.

removal of small pixel blocks strategy (see Fig. 12). Examining the Sheyang station cloud detection result on August 30 demonstrates the following (see Fig. 12).

1) Frequency filtering in time dimension removes spurious streaks and clear-air echo. In Fig. 12(a), the green area labeled F0 represents the region removed by time-dimensional frequency filtering. In addition to clear-air echo, what is removed is mainly the spurious streaks on the bottom of the cloud layer. In this study, it is believed that the main reason for this is that there are abnormal bumps at the bottom of the clouds mainly caused by the observation noise, and a better removal can be achieved by setting the filtering window in a reasonable way, ensuring the continuity of the layer.

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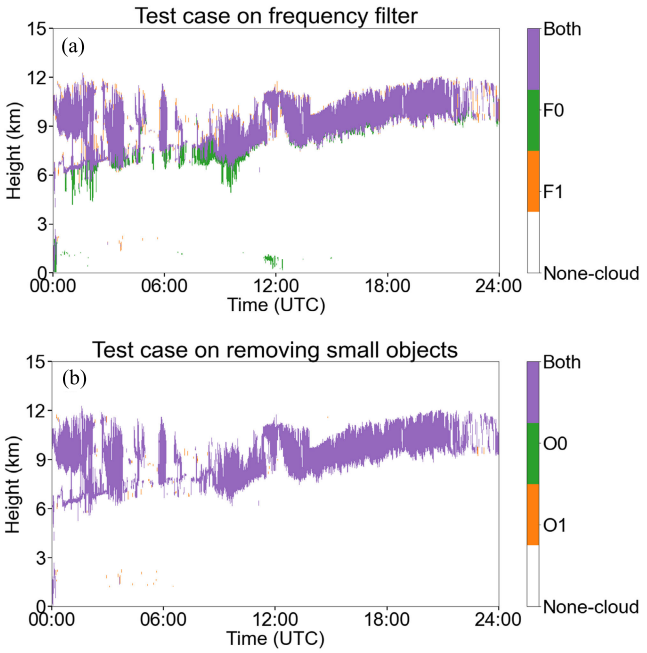


Fig. 12. (a) Comparison test on frequency filter strategy. Label F0 is the area that is only detected in the algorithm without frequency filter. F1 is the area that is only detected in the algorithm with frequency filter. (b) Comparison test case on removing small objects strategy. Label O0 is the area that is only detected in the algorithm where small objects are removed. O1 is the area that is only detected in the algorithm without removing small objects.

2) Fig. 12(b) shows a comparison test on the removal of small pixel blocks strategy. The orange O1 region represents the removed small objects, which are mainly clear-air echo. The strategy shows better suppression combined with the frequency number filtering strategy.

In summary, both strategies provide tangible improvements at reasonable computational expense.

C. Case 2: The Rainfall Identification Performance of the Algorithm

The proposed scheme utilizes Doppler velocity measurements and measures the difference between cloud base measured by radar and lidar observation for rainfall identification, as described in Section III. To evaluate reliability, data from September 12 encompassing light to heavy rain were analyzed (see Fig. 13).

During rainfall, larger raindrops, compared to cloud droplets, predominantly influence reflectivity. When the lidar signal is not fully attenuated [from 00:00 to 18:00 in Fig. 13(a)], the signal detected below the cloud layer is weaker than within the cloud but stronger than that from aerosols or the atmosphere. An increase in signal strength indicates a significant decrease in number concentration and a notable increase in particle size. The enhanced signal at the cloud base observed by the cloud radar also indicates a significant increase in particle size, consistent with the rainfall formation process.

Combined observations of Doppler velocity [see Fig. 14(b)], lidar signals [see Fig. 13(a)], and cloud radar signals [see Fig. 13(b)] around 4 km demonstrate the consistency among signal strength variations, particle size changes, and Doppler

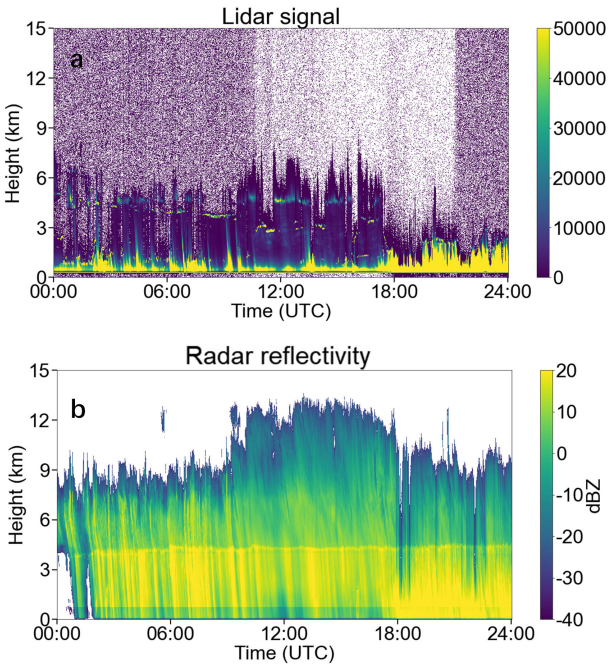


Fig. 13. Observation results on September 12. (a) lidar signal. (b) Radar reflectivity.

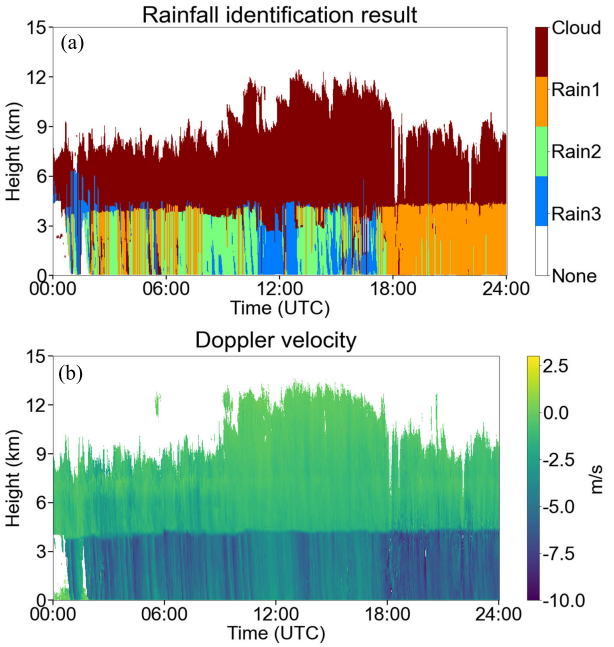


Fig. 14. (a) Rainfall identification result, label Cloud means the cloud detection result; orange label Rain1 means that rainfall identified by the Doppler velocity; green label Rain2 means that rainfall identified by both Doppler velocity and different cloud base; and blue label Rain3 means that rainfall identified by different cloud base only. (b) Doppler velocity observation.

velocity changes. The significant changes in particle size and Doppler velocity at this location validate the identification of the cloud base within this framework. In Fig. 13(a), the lidar shows strong attenuation below 5 km, while able to identify parts of the cloud base in drizzle due to attenuation from 00:00 to 17:00. In Fig. 13(b), a noticeable radar echo change occurs near 5 km during the whole day, indicating the melting layer around 5 km, where cloud droplets likely underwent phase change into falling raindrops. Ground observations showed

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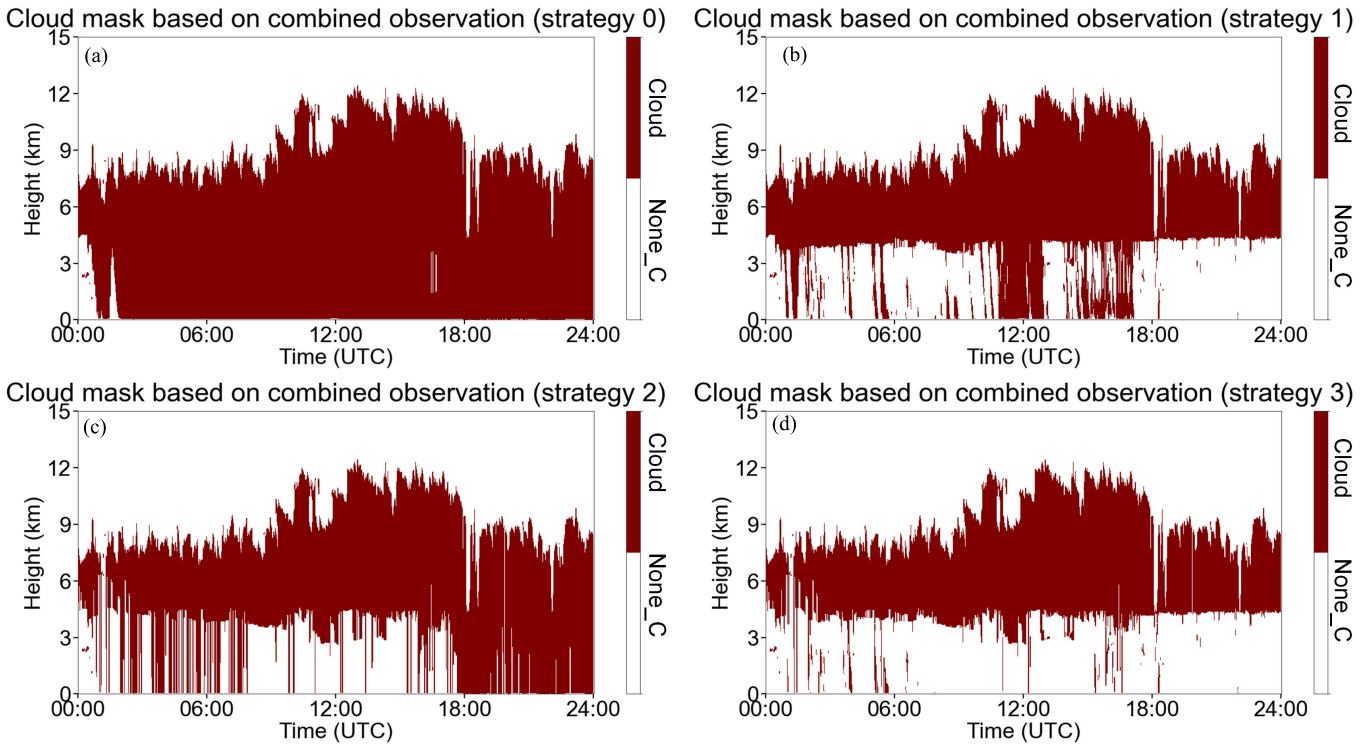


Fig. 15. Comparison on using different rainfall identification algorithms. (a) Strategy 0, do not identify rainfall. (b) Strategy 1, identify rainfall only based on Doppler velocity. (c) Strategy 2, identify rainfall only based on different cloud base detection. (d) Strategy 3, identify rainfall based on strategies 1 and 2.

light to heavy rain at Sheyang on September 12. In order to prove the effective rainfall identification ability of our algorithm, a test case is carried out (see Fig. 15).

As shown in Section III, two steps are used to identify rainfall. First, the Doppler velocity is used to detect rain droplets. Second, the different cloud bases as derived from lidar and radar are used to identify drizzle. Therefore, four rainfall identification strategies were compared in Fig. 15: strategy0 (S0) means the case of no rainfall removed, strategy1 (S1) means the case that only Doppler velocity is used for rain removal, strategy2 (S2) means the case that only cloud base difference is used to identify rainfall, and strategy3 (S3) means the case that we use the combined information of Doppler velocity and cloud base difference. Fig. 14 shows that S3 performs best overall, though S1 and S2 provide complementary information. Our analysis of data from a ground-based rain gauge reveals that the drizzle penetration of lidar complements the heavier rain from the Doppler velocities observation. If S3 is the reference, in this case, S1 and S2 have accuracies of 79.1% and 54.0%, respectively.

The rainfall identification result is shown in Fig. 14(a), where orange label Rain1 means rainfall identified by Doppler velocity, green label Rain2 means rainfall identified by both Doppler velocity and different cloud base, and blue label Rain3 means rainfall identified by different cloud base only. Fig. 14(a) and (b), from 18:00 to 24:00, demonstrates the Doppler velocity effectively indicates moderate-to-heavy rain. However, drizzle is more difficult to distinguish due to sparse small droplets [Fig. 14(b), from 11:00 to 18:00]. The reliable lidar cloud base aids identification in these cases (blue label in Fig. 14).

Time-series analysis over 128 days at Sheyang verifies the superior rainfall delineation of the composite algorithm. Compared to S3, S1 and S2 achieve only 58.6% and 53.2% detection ability, respectively.

D. Case 3: The Importance of Accurate Cloud Base During Rainfall

Another example is the observations from the Sheyang site on September 22, from 18:00 to 24:00.

In this case (see Fig. 16), starting from 19:00, the cloud base experienced a significant downward movement until it approached the melting layer, eventually leading to rainfall. During this period, the cloud base fluctuated between 4.3 and 4.1 km [see Fig. 16(c)]. These variations indicate that the conditions for raindrop formation are not entirely stable during precipitation. The pressure and temperature fluctuate with changes in rain intensity, cloud water content, and turbulence. Incorporating this information is valuable for understanding raindrop formation.

The proposed cloud identification scheme effectively captured the cloud base variations from before to during the rainfall. Given that raindrops can take from several minutes to hours to fall, accurately determining cloud base and cloud thickness information is crucial during heavy rainfall events. This information allows for accurate classification of the observed phenomena, enabling the use of different microphysical parameter retrieval schemes and facilitating effective estimation of cloud and rainwater paths. By assessing changes in water content, short-term surface precipitation can be accurately estimated, providing reliable short-term

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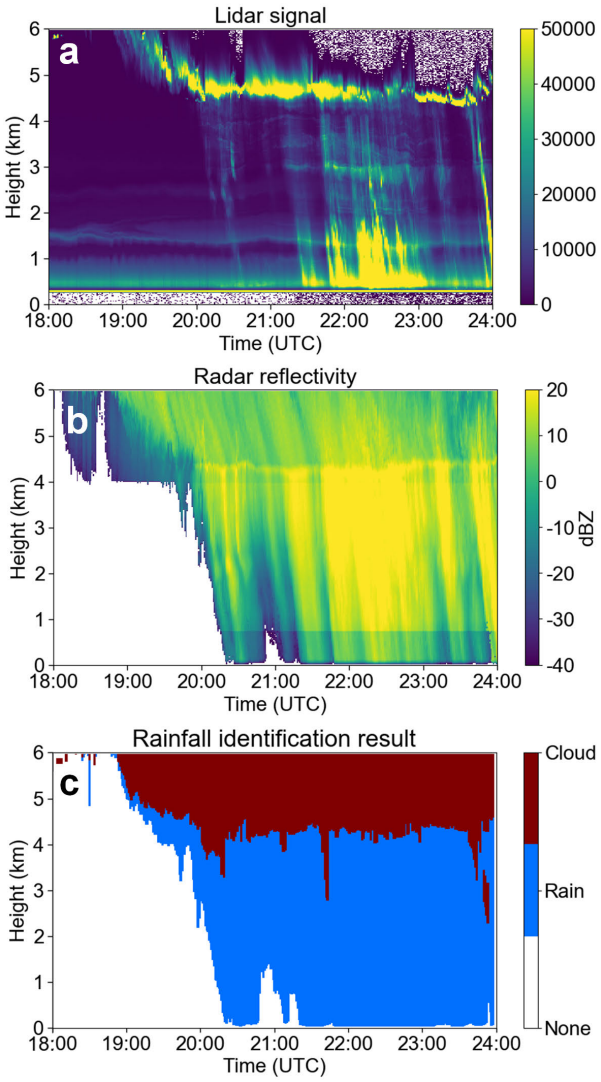


Fig. 16. Observation results on September 12. (a) Lidar signal. (b) Radar reflectivity. (c) Rainfall identification result.

rainfall forecasts

$$\mathrm{LWC} = (Z/0.0112)^{-1.16}\ \mathrm{g/m^3} \quad \text{for liquid cloud} \tag{10}$$

$$\mathrm{LWC} = 3.45 \times 10^{-3} \times Z^{0.571}\ \mathrm{g/m^3} \quad \text{for rainfall.} \tag{11}$$

Inaccurate cloud base information can lead to some rainfall being misclassified as water clouds during cloud water content retrieval, resulting in severe underestimation due to different Z–M relationships for various types. For example, comparing the raindrop water content calculation method by Brandes et al. [52] [formula (10)] with the cloud water content calculation method by Pujol et al. [37] [formula (11)], a misclassification at 10 dBZ could underestimate the water content by 97.1%. This discrepancy arises because raindrops and cloud droplets have significantly different numbers of concentrations and sizes, disrupting the empirical Z–M relationship.

During rainfall, the microwave radiometer's cloud base retrieval can result in an error of nearly 1 km [53]. Given a cloud base height variation between 3 and 6 km, such inaccuracies could lead to a 24.3%–48.5% error in estimating the rainwater path. This correction is particularly significant for data from platforms such as CloudSat that lack Doppler velocity observations, which can enhance the accuracy of cloud water content estimation.

V. Conclusion and Discussion

This study developed a composite cloud boundary detection method combining radar and lidar observations, obtaining precise cloud boundary by improving rainfall identification ability. The method effectively leverages the cloud penetration capability of radar and reliable cloud base detection ability of lidar. Lidar detected ice clouds to supplement cloud radar results by recognizing clouds radar missed. The algorithm considers the uncertainty in each cloud detection and transforms the binary detection problem into assigning cloud possibility. Reasonable threshold-insensitive approaches to determine cloud possibilities based on each instrument were developed. In addition, this study provides a detailed discussion on the thresholds used. Finally, small object removal and frequency filter were applied based on image processing concepts to effectively remove clear-air echo and spurious streaks, improving detection of low-attitude clouds. In time-series analysis, lidar alone detected only 41.1% of cloud pixels, while radar detected 93.2% pixels compared to using the composite method. This indicates that lidar detection effectively complemented radar and filled gaps. The algorithm was robust to threshold selection, supporting both composite and individual instrument cloud boundary detection.

This study demonstrates through observation and transmission equations that abrupt changes in lidar signals, cloud radar signals, and Doppler velocity observations are due to a sudden increase in particle size, distinguishing between raindrops and cloud droplets. The method also improved rainfall area detection by leveraging Doppler velocity and cloud base differences. Using this algorithm, rainfall area detection was enhanced by up to 53.2% compared to using Doppler velocity alone, which could only detect 58.6% of the rainfall area.

It is also important to acknowledge the presence of virga—fine streaks or wisps of precipitation that evaporate before reaching the Earth's surface—though they do not directly reach the ground, they still significantly affect microwave radiometer precipitation estimates and vertical thermal structures. Their occurrence frequency accounts for approximately 30% of precipitation. Virga precipitation is no longer treated as a separate category but is directly included in rain.

For the microwave radiometer's microphysical retrieval under precipitation conditions, identifying the cloud base during rainfall can bring an average improvement of 32.4%. However, the ambiguous distinction between cloud and rainfall introduces situations where rain may coexist with clouds. Further quantitative differentiation relies on aircraft measurement experiments and the retrieval of the physical properties of radar signals. In addition to Doppler velocity, spectral width should also be considered for detailed inversion. Utilizing the 2-D information from the Doppler spectrum distribution to quantify the cloud–rain relationship, and decomposing and identifying Doppler spectra generated by cloud–rain interactions, enhances the accuracy of feature identification and optimizes recognition effectiveness.

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A consistency analysis of cloud layer detection from time-series observations at the Sheyang station along with radiosonde data was also performed. Cloud bottom detection accuracy was found to be higher than cloud top detection. Statistical analysis of time-series results showed the composite cloud boundary detection method disagreed more with radiosonde for cloud base (overestimated by 855.1 m or 21.3%) than cloud top (underestimated by 551.2 m or 9.2%). However, when rainfall detection was excluded, the cloud base height was only underestimated by 228.8 m or 5.7%. This indicates that radiosonde method also lacked the ability to identify rainfall areas. The underestimation of cloud top was likely due to the continued response of the radiosonde humidity sensor while passing through cloud layers, reducing sensitivity.

This algorithm successfully accomplished precise cloud layer detection but had some limitations. First, the main methods were based on single profiles. Implementing algorithms using time-series data is reasonable and an important direction for future work. Second, misclassification errors remain, such as for extremely thin, small clouds, and thick clouds with small horizontal extent, which need improvement. Finally, cloud observation verification is challenging. Traditional radiosondes for radar–lidar consistency testing have limitations due to differences in observation methods, timing, and spacing compared to radar sampling instantaneously along fixed directions possibly introducing drift errors in radiosondes over ascend time. More reliable comparisons, such as using weather towers, drones, and other techniques, are needed in the future.

In conclusion, this study developed a composite cloud boundary detection method integrating lidar and radar observations. The method leveraged the strength of each instrument, addressed uncertainty, and filtered results to improve cloud and rainfall detection, providing more precise cloud boundary. While accomplishing the objectives, opportunities remain to reduce errors and validate the approach utilizing alternative observation strategies mentioned. Continued work refining algorithms and evaluation techniques will strengthen cloud observations and ultimately benefit weather forecasting and climate modeling applications. In addition, a comprehensive analysis of cloud distribution patterns across nationwide observation sites will support the evaluation of CH results from EarthCARE and Fengyun satellites.

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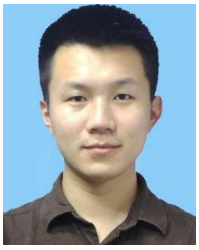
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